

Title: To design and develop a user sign up and sign in page for an e-commerce Application Online Auctioning System

Objective

- To design the Login and Registration pages for an Online Auctioning System.
- To implement client-side validation using JavaScript.
- To ensure users enter valid information before submitting the form.
- To improve the reliability of user input through form validation.

Theory

Form validation is an essential part of web application development that ensures users provide correct and complete information before a form is submitted. In an Online Auctioning System, both the Login and Registration pages require validation to prevent invalid or incomplete data from being processed.

HTML is used to create the form structure, CSS is used to design the user interface, and JavaScript is used to perform client-side validation. The validation checks whether required fields are filled, verifies the email format, ensures the password meets the minimum length requirement, confirms that the password and confirm password fields match, and ensures that a user role is selected.

Technologies Used:

- HTML5 – Form structure
- CSS3 – Page styling
- JavaScript – Form validation

Source Code:

1. Required Field Validation

```
if (!fullName || !email || !password || !confirmPassword || !role) {  
    setMessage(message, "All fields are required.", "error");  
    return;  
}
```

2. Email Validation

```
const EMAIL_REGEX = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

function validateEmail(email) {

    return EMAIL_REGEX.test(email.trim());
}

if (!validateEmail(email)) {
    setMessage(message, "Please enter a valid email address.", "error");
    return;
}
```

3. Password Length Validation

```
if (password.length < 8) {
    setMessage(message, "Password must contain at least 8 characters.", "error");
    return;
}
```

4. Confirm Password Validation

```
if (password !== confirmPassword) {
    setMessage(message, "Passwords do not match.", "error");
    return;
}
```

5. Login Field Validation

```
if (!email || !password || !role) {
    setMessage(message, "All fields are required.", "error");
    return;
}
```

Output

ONLINE AUCTION SYSTEM

 **AuctionHub**

Log in to browse live auctions, manage listings, and keep track of your bids or items.

Email

Password

User Role

☐ Buyer

☐ Seller

Login

Don't have an account? [Sign Up](#)

CREATE ACCOUNT

 **AuctionHub**

Join AuctionHub to place bids as a buyer or publish items as a seller.

Full Name

Email

Password

Confirm Password

User Role

☐ Buyer

☐ Seller

Sign Up

Already have an account? [Login](#)

Conclusion

In this lab, the Login and Registration pages for the Online Auctioning System were successfully developed with client-side form validation using JavaScript. The validation ensures that all required fields are completed, email addresses are entered in the correct format, passwords meet the minimum length requirement, and the password confirmation matches. These validations help improve data accuracy and provide a better user experience before moving on to database integration and authentication in future labs.